TerraOS

AI-Powered Soil Intelligence Platform

Smart Soil. Better Harvest. For Every Indian Farmer.

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1. Project Overview

TerraOS is an AI-powered soil intelligence platform designed to transform basic soil test values — Nitrogen (N), Phosphorus (P), Potassium (K), and pH — into a comprehensive agricultural intelligence report. The entire analysis is completed in under a couple of minutes and is powered by a 5-agent Amazon Bedrock pipeline running on fully serverless AWS infrastructure.

The platform is purpose-built for Indian farmers, offering multilingual support, zero lab requirements, and instant actionable recommendations — including carbon credit eligibility — at no infrastructure cost to the farmer.

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2. Problem Statement

Indian agriculture faces a critical set of challenges that directly impact farmer productivity and income:

Challenge	Scale
Indian farmers affected	140 million+
Yield lost to poor soil management	15–40%
Farmers without soil testing access	60%
Lab test cost and wait time	$6 - $22 approx. + 2–4 weeks
Carbon credit awareness among eligible farmers	< 1%

The vast majority of Indian farmers lack access to affordable, timely soil analysis — and almost none are aware of their eligibility for carbon credit income programs.

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3. Solution

TerraOS provides instant, affordable, and multilingual soil intelligence with no laboratory required. A farmer inputs just four basic soil parameters and receives a full intelligence report within minutes, covering:

- Soil health score (0–100) and grade (A–F)
- NPK and micronutrient status with fertilizer recommendations
- Priority action items and crop recommendations
- Carbon credit eligibility and potential income
- Historical soil trend analysis
- Output available in 7 Indian languages

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4. System Architecture

TerraOS follows a fully serverless, event-driven architecture on AWS, deployed in the ap-south-1 (Mumbai) region. The flow from farmer input to final report proceeds through the following stages:

4.1 Request Flow

- The farmer submits soil data via the React web application.
- The request is routed through Amazon API Gateway with Cognito JWT authorization.
- A Lambda function (api-handler) writes raw input to S3 Bronze layer, organized by farmer ID and timestamp.
- An S3 event triggers the input-trigger Lambda, which invokes AWS Glue Job 1 (Bronze Cleaner) to validate and clean the data.
- AWS Step Functions orchestrates the multi-agent AI pipeline in sequence and parallel.
- The final processed report is stored in S3 Gold layer and indexed in DynamoDB for fast retrieval.
- The report-fetcher Lambda serves the completed report back to the frontend.

4.2 Data Lake Layers

Layer	S3 Bucket	Contents
Bronze	terraos.bucket.bronze	Raw farmer input + Glue-cleaned data
Silver	terraos.bucket.silver	NPK analysis + suggestion report
Gold	terraos.bucket.gold	Final minified report (React-ready)

4.3 Glue ETL Pipeline

Three AWS Glue Python Shell jobs process data through each lake layer:

- Glue Job 1 — Bronze Cleaner: Validates null values and checks ranges (N: 0–600, P: 0–200, K: 0–600, pH: 0–14). Rejects invalid entries without applying defaults.

- Glue Job 2 — Silver Filter: Strips full NPK analysis and retains only historical fields, soil score, and grade for trend comparison.
- Glue Job 3 — Gold Minifier: Removes raw Bedrock output, deduplicates lists, and produces a clean React-ready JSON format.

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5. AI Agent Pipeline

The core intelligence of TerraOS is delivered through five specialist AI agents powered by Amazon Bedrock (model: amazon.nova-pro-v1:0, region: us-east-1), orchestrated via AWS Step Functions.

Agent 1 — NPK Agent (Parallel Execution)

- Analyzes Nitrogen, Phosphorus, and Potassium levels against optimal ranges
- Assesses soil pH
- Generates an NPK score and specific fertilizer recommendations

Agent 2 — Nutrient Agent (Parallel Execution)

- Infers micronutrient levels, organic carbon, and cation exchange capacity (CEC) from the four base inputs
- Evaluates microbial health and broader soil ecosystem indicators

Agent 3 — Suggestion Agent (Sequential)

- Combines outputs from NPK and Nutrient Agents
- Computes final soil health score (0–100) and grade (A–F)
- Generates priority action items and crop recommendations
- Writes output to S3 Silver layer and triggers Glue Job 2

Agent 4 — Historical Agent (Sequential)

- Reads all past reports for the farmer from S3 Silver
- Identifies soil trends (improving, declining, stable) across N, P, and K
- Adjusts recommendations based on historical trajectory

Agent 5 — Carbon Agent (Sequential)

- Calculates CO_2 sequestration potential based on soil health
- Determines eligibility for Gold Standard, VCS, and Plan Vivo carbon credit programs
- Writes final report to S3 Gold layer and records metadata in DynamoDB
- Triggers Glue Job 3 for final minification

Agent Retry Strategy

Each agent implements exponential backoff with jitter across up to 4 attempts (wait intervals: 2s, 4s, 8s). On continued failure, AWS Step Functions catches the exception and routes to an error state.

6. AWS Services

Service	Purpose	Configuration
Amazon Bedrock	5 AI agents	amazon.nova-pro-v1:0 · us-east-1
AWS Lambda	8 serverless functions	Python 3.12 · 128–256 MB
AWS Step Functions	Pipeline orchestration	Express workflow
Amazon S3	Data lake (3 buckets)	Bronze / Silver / Gold
AWS Glue	ETL data cleaning	Python Shell · 0.0625 DPU
Amazon DynamoDB	Report metadata storage	On-demand · ap-south-1
Amazon Cognito	Authentication	Email OTP + JWT
Amazon API Gateway	REST API endpoint	Cognito JWT Authorizer
AWS Amplify	Frontend hosting	React build
Amazon Translate	Multilingual support	6 Indian languages, real-time
Amazon CloudWatch	Logging & monitoring	All Lambda + Glue functions

Lambda Function Timeouts

Function	Timeout	Memory	Reason
api-handler	30s	128 MB	S3 + DynamoDB write
input-trigger	5 min	128 MB	Waits for Glue completion
npk-agent	2 min	128 MB	Bedrock + retries
nutrient-agent	2 min	128 MB	Bedrock + retries
suggestion-agent	3 min	128 MB	Bedrock + S3 + Glue trigger
historical-agent	3 min	128 MB	S3 list + Bedrock
carbon-agent	3 min	128 MB	Bedrock + S3 + DynamoDB + Glue
report-fetcher	30s	128 MB	S3 list + reads

7. Frontend Application

The frontend is a React + Vite single-page application hosted on AWS Amplify. It communicates with the backend via API Gateway and supports real-time multilingual translation using Amazon Translate.

Key Pages

- Home Page — Soil score summary dashboard for the farmer
- Analyze Page — Soil input form that calls POST /analyze
- Loading Page — Polling screen while AI agents process
- Report Page — Full soil intelligence report view
- History Page — All past reports for the farmer
- Profile Page — Farmer profile, grade, and report count
- Login Page — Cognito email OTP login and signup

Key Components

- SoilScoreGauge — Animated circular gauge showing score 0–100
- CarbonCard — Carbon credit eligibility display
- NutrientCard — NPK and micronutrient status
- PracticeAccordion — Expandable regenerative practices list
- TopNav — Top navigation with language switcher

Language Support

The platform supports 6 languages: English, Hindi (हिन्दी), Marathi (मराठी), Tamil (தமிழ்), Telugu (తెలుగు) and Bengali (বাংলা). Translations are cached in memory after the first fetch.

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8. API Reference

Submit Soil Analysis

POST /analyze | Authorization: Bearer {cognito_jwt}

Accepts: nitrogen, phosphorus, potassium, ph, crop_type, location, soil_type, farm_size_acres

Returns: farmer_id, report_id, timestamp, estimated_time

Fetch Latest Report

GET /report/{farmerId}?latest=true | Authorization: Bearer {cognito_jwt}

Returns: status (complete/processing), and full report JSON including soil_health, npk, carbon, and historical sections when complete.

Fetch All Reports

GET /report/{farmerId} | Authorization: Bearer {cognito_jwt}

Returns all historical reports for the given farmer ID.

9. Data Sovereignty & Security

- All farmer data is stored under isolated S3 key prefixes using farmer_id — no cross-farmer data contamination is possible.
- Authentication is enforced at the API Gateway level via Cognito JWT tokens.
- All Bedrock API credentials are managed as per-Lambda environment variables (never hardcoded).
- CloudWatch logging is enabled on all Lambda functions and Glue jobs for full auditability.

10. Key Innovations

Innovation	Description
Zero-Lab Analysis	AI infers micronutrients, organic carbon, and CEC from just 4 basic soil parameters — no lab equipment needed.
Carbon Economy Access	First platform connecting Indian farmers to Gold Standard, VCS, and Plan Vivo carbon credit programs.
Multilingual by Design	Full report output in 6 Indian languages with real-time translation via Amazon Translate.
Regenerative Focus	Recommendations prioritize long-term soil health improvement over quick chemical fixes.
Serverless & Affordable	Zero infrastructure cost for the farmer — entirely event-driven on AWS serverless services.
Data Sovereignty	Per-farmer S3 isolation by farmer_id prefix ensures strict data separation.

11. Target Impact

- 140 million+ Indian farmers as the addressable audience
- 60% currently without any access to soil testing
- Carbon credit eligibility identified and communicated per farmer
- Potential for 15–40% yield improvement through proper soil management

- Near-zero cost per analysis due to fully serverless infrastructure

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